C# OOP SQL

Project

Cairo Coding School

Project
Quiz / Exam System
Overview

- is a software application used to create, manage, and evaluate quizzes for students.
- It allows teachers to create questions and quizzes, and the system automatically evaluates student performance.
- The system is built using OOP principles and uses a SQL database for data storage.

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Features

Admin

- Login
  - Enter Email and Password.
  - Admin credentials are not created داخل النظام (no function/page for adding admin).
- Add Courses
  - Add all course details:
  - Name
  - Number of Lessons
  - Category
  - Duration
  - The system should know which Admin created the course.

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Features

Admin

- View All Courses
  - Display the following details for each course:
    - Course Name
    - Teacher Name
    - Duration
    - Number of Lessons
    - Category
    - Number of Students
    - Number of Quizzes
    - List of Quiz Names
    - List of Student Names

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Features

Admin

- View All Teachers
  - Display the following details for each teacher:
    - Teacher Name
    - Email
    - Title
    - Number of Courses
    - Number of Quizzes Created
    - List of Course Names with their Categories
    - List of Quiz Names

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Features

Admin

- View All Students
  - Display the following details for each student:
    - Student Name
    - Email
    - Grade
    - Number of Courses
    - Number of Quizzes Taken
    - List of Course Names
    - List of Quiz Names

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Teacher

- Register and log in
- Assign themselves to a course
- View assigned courses with details
- View unassigned courses without details
- A teacher can have multiple courses
- A course can be assigned to only one teacher
- Can release a course if no students have taken it

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Teacher

- Create quizzes for a course
- Edit quizzes (add/remove questions) if no students have taken them
- A quiz can contain:
  - True / False
  - Multiple Choice
  - Short Answer

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Teacher

- View unassigned course details:
  - Name, category, duration
- View assigned course details:
  - Name, category, duration
  - Number of lessons and quizzes
- View quiz details:
  - Quiz name
  - Duration
  - Number of students who took it
  - All questions

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Student

- View all courses
  - Display the following details for each course:
    - Course Name
    - Teacher Name
    - Duration
    - Number of Lessons
    - Category
    - Number of Quizzes
    - List of Quiz Names

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Student

- **can** Enroll in a course if course have a teacher
- View entrolled course details and quizzes
- Take quiz within a specific duration
  - If time finishes → quiz ends and score is shown
  - System evaluates answers and calculates score

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Student

- **View all quizzes taken by the student**
- View specific quiz details:
  - Quiz name
  - Course name
  - Quiz Duration
  - Duration
  - Score
  - Questions
  - Student answers
  - Correct answers

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Objective

- Create a Console Application that implements the full Quiz / Exam System functionality described previously.
- The Main method should:
  - Must NOT contain business logic
- Apply all OOP concepts discussed in sessions
- Think about missing requirements
- Organize code into multiple classes (no single large class) in different files

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Restrictions

- Do NOT use magic numbers
- Do NOT use hardcoded strings (magic strings)
- Do NOT rely on external help (e.g., ChatGPT, tutorials, or libraries not discussed)
- Do NOT use concepts or tools that were NOT covered in the sessions

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Implementation Notes

- Create a clear class design
  - each class has one responsibility
- Use interfaces to define common behaviors
- Keep the system modular and extendable

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Objective

- Design and implement a complete SQL database for the Quiz / Exam System.
- Create all required tables for the system
- Define clear relationships between tables
- Apply all necessary SQL constraints
- Handle all relationships correctly
- Insert at least 20 records in EACH table
- Data should be:
  - منطقي (logical)
  - مترابط (related correctly باستخدام Foreign Keys)

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- Return all course names with their teacher names and category
- Return all quiz names with the number of questions in each quiz
- Return all questions for a specific quiz
- Return all student names and the course name they are enrolled in for a specific course
- Return the number of quizzes in a specific course
- Return all courses with their teacher, including courses that have no teacher assigned

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Sql

- Update course duration
- Delete a quiz
  - If there are no students who have taken this quiz
  - Also delete all related data
- Calculate the average quiz duration
- Calculate the average student score per quiz
- Show student name, student answer, correct answer, and score per quiz

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Sql

- Get courses with more than 5 students
- Get courses that have quizzes
- Add Admin details
- Generate a report that shows how a specific student performed across quizzes and courses.
    - Student Name , Course Name , Quiz Name , Total Score , and Result (Pass / Fail)

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- Analyze the query requirements carefully.
- Based on the requirements, determine the appropriate SQL object type:
  - View
  - Scalar Function
  - Table Function
  - Stored Procedure
- Important:
  - Do not include any queries that do not match the given requirements. Only format and describe queries that align with the requirements.

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Each student should send:

- Console Application
- A SQL script to create the database structure (tables, columns, and constraints)
- A SQL script to create Views, Scalar Functions, Table Functions, and Stored Procedures



If you have any questions or need assistance with anything, feel free to ask us. We are here to help you

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